

MONTEVIDEO CARRASCO, Uruguay

WMO: 865800

Lat: 34.8333S

Lon: 56.0117W

Elev: 34

StdP: 100.92

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.5	3.0	-2.1	3.2	6.6	-0.9	3.5	6.8	12.1	9.2	10.7	10.7	1.9	320	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	31.9	21.8	30.0	21.5	28.5	21.2	24.3	28.6	23.4	27.2	22.6	26.1	5.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	17.9	26.3	22.1	16.9	25.2	21.5	16.2	24.5	73.6	28.5	69.9	26.9	67.1	26.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.5	8.5	-1.2	35.6	1.6	1.4	-2.3	36.6	-3.2	37.4	-4.1	38.2	-5.2	39.2		
			-1.6	26.2	1.4	1.2	-2.6	27.0	-3.4	27.7	-4.2	28.4	-5.2	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	22.4	22.0	20.3	17.3	14.1	11.1	10.5	11.9	13.1	15.7	18.4	21.0	(6)
(7)		DBStd	5.22	2.73	2.71	2.66	3.04	3.19	3.39	3.65	3.73	3.22	2.95	3.13	3.10	(7)
(8)		HDD10.0	91	0	0	0	0	3	23	38	19	7	1	0	0	(8)
(9)		HDD18.3	1175	2	3	11	55	137	218	246	204	160	91	37	11	(9)
(10)		CDD10.0	2447	385	336	320	218	129	57	52	78	101	178	252	339	(10)
(11)		CDD18.3	489	129	107	73	23	4	2	2	4	4	11	38	92	(11)
(12)		CDH23.3	3022	872	618	333	89	20	5	7	33	25	77	284	658	(12)
(13)		CDH26.7	827	263	164	66	14	1	0	0	8	4	15	76	216	(13)
(14)	Wind	WSAvg	4.3	4.6	4.4	4.2	4.0	3.7	3.9	4.1	4.2	4.6	4.6	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	1044	88	92	101	105	71	82	80	73	79	95	93	85	(15)
(16)		PrecMax	1666	292	244	460	275	241	221	195	236	197	213	216	273	(16)
(17)		PrecMin	576	2	5	12	0	4	6	12	8	11	7	7	5	(17)
(18)		PrecStd	229	67	56	89	69	45	50	47	51	47	54	55	62	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	34.5	33.7	31.0	28.5	26.0	23.8	24.2	27.2	26.2	28.8	31.5	33.8	(19)	
(20)		MCWB	22.7	23.3	22.2	20.7	19.1	19.3	18.5	19.6	19.6	19.5	20.8	21.6	(20)	
(21)		DB	31.5	30.2	28.5	25.8	22.9	20.2	20.2	23.0	23.1	25.3	29.0	31.1	(21)	
(22)		MCWB	22.1	22.5	21.1	19.4	17.7	16.9	16.2	17.2	17.0	18.1	19.9	21.1	(22)	
(23)		DB	29.3	28.7	26.8	23.9	20.6	18.1	18.0	20.0	20.9	23.0	26.8	29.1	(23)	
(24)		MCWB	21.7	22.3	20.8	18.8	16.9	15.3	15.0	15.7	15.4	16.7	19.1	20.6	(24)	
(25)		DB	27.8	27.0	25.0	22.1	18.9	16.2	16.0	17.9	18.7	21.0	24.2	27.0	(25)	
(26)		MCWB	21.6	21.3	20.6	18.6	16.3	13.8	13.5	14.4	14.5	16.4	18.2	20.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	25.5	25.5	24.6	22.7	20.4	20.1	18.9	20.3	21.0	21.5	22.9	24.6	(27)	
(28)		MCDWB	30.3	29.9	28.2	25.9	24.2	22.4	23.4	25.9	24.9	26.3	28.0	30.1	(28)	
(29)		WB	24.1	24.2	23.1	21.2	18.7	18.1	17.2	18.4	18.4	19.6	21.3	23.1	(29)	
(30)		MCDWB	28.5	28.2	26.0	23.6	21.3	19.9	19.6	22.0	21.6	22.9	25.7	27.7	(30)	
(31)		WB	23.2	23.2	22.1	20.3	17.7	16.2	15.7	16.7	16.7	18.5	20.3	22.1	(31)	
(32)		MCDWB	27.0	26.6	24.9	22.5	19.6	17.6	17.3	19.2	19.2	21.2	24.5	26.2	(32)	
(33)		WB	22.5	22.4	21.4	19.3	16.7	14.7	14.0	15.1	15.2	17.4	19.3	21.2	(33)	
(34)		MCDWB	26.0	25.6	23.8	21.5	18.4	16.0	15.5	17.3	17.7	20.2	23.0	25.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.2	8.5	8.4	8.5	8.1	8.3	8.1	8.7	8.4	8.4	9.5	9.6	(35)	
(36)		5% DB	12.5	11.5	11.4	11.1	10.7	10.0	10.4	12.3	12.5	12.7	13.7	13.7	(36)	
(37)		MCWBR	6.2	5.7	5.9	6.1	6.3	6.7	6.6	7.3	7.2	7.0	6.8	6.5	(37)	
(38)		5% WB	9.8	9.2	8.6	8.1	8.2	8.0	8.7	10.6	9.8	9.7	11.0	10.4	(38)	
(39)		MCWBR	5.7	5.4	5.2	5.4	5.6	6.3	6.4	7.1	6.8	6.3	6.5	5.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.394	0.390	0.369	0.365	0.357	0.360	0.361	0.442	0.454	0.400	0.383	0.384	(40)	
(41)		taud	2.424	2.456	2.502	2.460	2.451	2.430	2.396	2.109	2.096	2.326	2.405	2.427	(41)	
(42)		Ebn at Noon	942	924	907	848	795	760	779	747	804	904	947	956	(42)	
(43)		Edh at Noon	123	115	102	94	84	81	87	132	150	129	124	123	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.33	6.28	5.26	3.82	2.65	2.22	2.32	3.09	4.29	5.61	6.85	7.53	(44)	
(45)		RadStd	0.49	0.58	0.46	0.41	0.19	0.21	0.23	0.31	0.31	0.49	0.54	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	+0.85	N/A	+0.81	+0.55	N/A	N/A	-127	+166	+83
(47)	Regional (6 neighbors)	+0.48	+0.70	N/A	+0.69	+0.63	N/A	N/A	N/A	+125	+67

Nomenclature: See separate page